

School of Science, Computing and Engineering Technologies



COS10025

Technology in an Indigenous Context Project

Final project reflection report

Project Title: E-Commerce Platforms for Traditional Handicrafts from Southern VietNam

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[Acknowledgement of Country]

[NOTE TO STUDENTS: Before submitting, remove and/or action with your own words all help instructions enclosed within square brackets [...] including this note.]

Declaration

I declare that this report is my individual work. I have not copied from any other student's work or from any other source except where due acknowledgment is made explicitly in the text, nor has any part of this submission been written for me by another person.

Signature:

A handwritten signature in dark ink, appearing to read "Truong Ngoc Gia Hieu", written over a horizontal line.

Part A:

Introduction (Project Description)

First and foremost, thousands of artisan villages and communities are widely distributed across Southern VietNam which not only showcase unique products and preserve cultural heritages but also display traditional artifacts to international tourists. However, due to the advancement of technological revolution, these historic handicrafts faces significant challenges nowadays including limited digital literacy and market reach, poor online sales access, hurdles in secure payment, and ability to sustain livelihoods. Because of following reasons, the Request for Proposal (RFP) organization decides to launch the project named “E-Commerce Platforms for Traditional Handicrafts across Southern Vietnam” that seeks innovative solutions to address recommended problems by empowering local artisans, improving life quality, and ensuring the long-term viability of their invaluable cultural contributions.

The final project reflection report indicates a detailed plan for the project, including methodologies, solutions, strategies, and a detailed roadmap for implementation, demonstrating a clear commitment to achieving impactful and sustainable outcomes for the artisan communities involved. Additionally, the report illustrates the critical alignment of the project with fair trade principles and its complicated approach to ensuring the long-term economic empowerment and cultural preservation of these traditional handicraft-producing groups. Specifically, participantinh in this fascinating project offers students to explore distinct cultural insights across Southern Vietnam and improves soft skills in e-commerce and digital marketing, and contribute to the preservation of traditional crafts while fostering sustainable economic growth within the artisan communities.

Part B

2.1 Project reflection

Due to the project specification, Hieu, as the team leader, decided to initilize a careful and detailed plan that defined individual roles and responsibilities, and set achievable goals, guiding the team's efforts throughout the duration of the project. Particularly, Hieu exceled in organizing and monitoring crucial meetings and keeping track of deadlines and assignments, ensuring team still maintain workflow. While Phuc signiificantly proved to be proficient in creating visually appealing mockups that served as a foundation for the coding phase and ensured that the final product would be user-friendly and aligned with our project goals, Hoang was outstanding in searching, analyzing, and gathering important information to determine which province would be chosen for the following phases. As a result, the team's collaboarative efforts and effective planning led to a successful project outcome that met directly all objectives and satisfied stakeholder expectations. During the project, team members not only discovered useful knowledge about different cultures and customs in each region but also enhanced soft skills including communication, teamwork, and problem-solving which are extremely useful for academic performance and future career development.

2.2 Group Work Strategies

During the project workflow, team members utilized powerful tools and applications such as Google Docs and AI Assistant, allowing team to gain valuable insights into fostering collaboration and efficiency throughout the project lifecycle. Firstly, my team GAMBLE utilizes Google Docs platform for organizing ideas, taking notes, documenting research findings, and collaboratively refining project concepts through real-time editing and commenting features proved invaluable. Secondly, using AI Assistants enabled team to streamline processes, boosted decision-making, provided quick access to relevant information, offered suggestions for improvement, and helped analyze data effectively. Consequently, by combining these integration of technologies into the project not only worked more efficiently and creatively throughout the project lifecycle but also enriched the quality of discussions and outcomes.

Although team had used appropriate strategies and methodologies, team dealt with several difficulties include coordinating schedules for meetings due to external factors, managing different levels of familiarity with the tools, and ensuring consistent communication among all members. Thus, these challenges frequently required team additional efforts to maintain alignment and focus on project goals. Due to external factors, team increased meeting frequency and review previous activities by implementing regular check-ins and feedback sessions to ensure everyone was on the same page and to foster a supportive environment for collaboration. Additionally, Hieu - team leader, frequently reflected on the team's progress and timeline, encouraging open discussions about what worked well and what could be improved. Thus, this activity not only helped identify areas for growth but also reinforced a culture of continuous learning and adaptation within the team.

Following each reflective process in each phase in the project workflow, team leader realized several perspectives for improvement in future collaborations, notably the proactive communication. In details, team members often waited for others to initiate discussions rather than taking the lead myself that could help in clarifying tasks and expectations more efficiently. On the positive side, team demonstrated outstanding organization, particularly in how meetings were structured. Specifically, a detailed and carefully plan was generated, enabling all members within team to contribute efforts and ideas into the project and ensuring everyone keep up with the timeline. However, the situation above revealed that there were instances where some team members felt hesitant to voice their opinions and suggestions, highlighting the need for creating a more open and encouraging atmosphere. In summary, fostering proactive communication and an inclusive environment will be essential for improving collaboration and ensuring that all team members feel valued and empowered to contribute ideas in future projects.

2.2.1 Team Organisation

Based on the project requirements, Hieu, team leader of group, launched a clear, detailed, and carefully preparation that highlighted specific objectives,

timelines, and responsibilities for each team member, ensuring that everyone was aligned and equipped to contribute effectively to the project's success. Moreover, Hieu also implemented regular check check-ins and progress updates to monitor the team's advancement towards the established goals, leading not only boost accountability but also allow for timely adjustments to the plan as needed, ensuring that any challenges could be addressed promptly. Consequently, generating a dynamic environment within group during the project offers great chances for team members to showcase opinions and ideas that improve overall effectiveness of the project. Moreover, the powerful preparation and ongoing assist contributed to the team's ability to meet deadlines and achieve high-quality specifications.

2.2.2 Meetings

During the project, Hieu frequently held several meetings per week by using Google Meeting and Notepad which are highly recommended for effective communication between team members and taking notes. Moreover, these consistent and well-structured meetings ensured that all team members remained aligned on evolving project goals and individual responsibilities throughout the project lifecycle. Notably, team members had ability to share real-time feedback and contributed their efforts to the project within comfortable and confidence environment, boosting collaboration and production between members.

2.2.3 Delivery of the Project Design Ideas

In team GAMBLE, Phuc showcased outstanding skill in the delivery of the project design ideas that was proficient in generating visually appealing mockups that served as a foundational element for the subsequent coding phase. As a result, Phuc guaranteed that the final product would be user-friendly and closely aligned with the overarching project goals.

2.2.3 Delivery of the Innovation Concept

During the workflow, the delivery of our innovation concept was significantly strengthened by the team's collaborative approach and the clear roles established from the outset. Specifically, Hoang's comprehensive market research, for instance, provided the critical data needed to shape and refine our innovative solutions, ensuring they were both novel and relevant to the target communities. Therefore, this collaborative ideation, supported by Google Docs for immediate feedback, enabled a robust and well-articulated innovation concept.

2.2.4 Delivery of the Final Presentation

The final presentation represents team's organized efforts and individual strengths. Therefore, Hieu's excelled in organizing crucial meetings and meticulously tracking deadlines played a vital role in ensuring that all presentation components were developed and integrated seamlessly. Furthermore, Phuc's skill in generating visually appealing mockups significantly improved the presentation's impact, effectively conveying our project's scope and achievements to the audience. Consequently, this combination of strong leadership, detailed preparation, and impactful visual communication allowed us to deliver a comprehensive and compelling final presentation.

2.3 Individual Work Reflection

This section illustrates a comprehensive reflection on the project named “E-Commerce Platforms for Traditional Handicrafts from Southern Vietnam”. Specifically, this part details my specific roles, responsibility, and tasks on each project phase, elaborates on how my efforts directly contributed to the group's overall success, and concludes with key achievements and recommendations for future collaborative endeavors.

2.3.1 First Phase

According to the first phase of the project requirements, as the team leader, Hieu launched an important meeting which ensures a comprehensive and culturally sensitive understanding of the artisan communities' needs. Furthermore, this activity involved guiding the team in conducting extensive consultations to identify market access challenges, digital literacy levels, and preferences for the e-commerce platforms. After finishing the meeting, Hieu launched a detailed plan and led the collaborative development of the overall platform strategy, including defining technical requirements and integrating fair trade principles.

2.3.2 Second Phase

Next, during the second phase, my team decided to utilize the Data Flow Diagram (DFD) which visually represented the flow of information through the e-commerce platform, allowing team members to meticulously plan and optimize data processes for both artisans and customers. Consequently, this approach offered team identify key data inputs, outputs, processes, and storage points within the system. For instance, for the design of the DFD for the User/Customer Interface, team GAMBLE meticulously mapped how customer data (inputs like login credentials, product selections, and shipping details) would flow through processes such as account creation, Browse, adding to cart, checkout, and order confirmation. Moreover, the DFD designs detailed mapping ensured a logical and efficient flow of information, informing the development of an intuitive and user-friendly interface that catered to diverse digital literacy levels, while also highlighting critical points for data security and integrity.

Based on the DFD design for the User/Customer, Artisan Community, and Administrative platforms, team members started to dive into the coding progress that involved translating these detailed blueprints into tangible, functional software. Firstly, for the User/Customer platform, team members developed a responsive and intuitive front-end, ensuring seamless navigation for Browse, product selection, and a secure checkout process. Secondly, my team focused on robust backend functionalities for the Artisan Community Interface by empowering artisans with easy-to-use tools for product listing, inventory management, order processing, and communication, prioritizing simplicity and accessibility given varying digital literacy levels. Lastly, the Administrative platform's coding determined on creating a complicated system for overall platform management, including data analytics, user management, and content moderation, all while adhering to best practices in web development and security to build a robust and secure e-commerce ecosystem.

2.3.3 Third Phase

Following the successful backend and frontend implementation on the second phase, based on the project requirements and leadership of team leader, team GAMBLE prioritized the development of culturally appropriate and easy-to-understand training materials including comprehensive video tutorials, detailed step-by-step guides, concise handouts, and an extensive FAQ section, all translated into relevant local languages. Notably, the content covered necessary various topics such as effective product photography, crafting compelling product descriptions, seamless listing creation, efficient inventory management, streamlined order processing, and best practices for customer service. Thus, this solution ensured maximum accessibility and continuous learning, these training resources will be integrated into a dedicated 'Artisan Community Platform'. Subsequently, team GAMBLE conducted hands-on training workshops for artisan communities across Southern Vietnam which were designed for small groups to allow for personalized support and direct engagement, addressing the specific digital literacy levels and unique challenges faced by each community. For example, team members proposed a two-day intensive workshop model, starting with an introductory session. Particularly, on day-one, team proposed a two-day intensive workshop model, starting with an introductory session to explain the project's vision, potential benefits, and importance to the local economy. In addition, on second day, team focused on practical, step-by-step instruction on navigating the e-commerce platform's interface, covering categories like Logistics, Trade, Support, Collaboration, Search Bar, Resources, Training, and News, providing live demonstrations, observing artisans as they practice, answering their questions in real-time, and collecting immediate feedback to refine our training approach.

2.3.4 Fourth Phase

Upon successful completion of artisan training and onboarding on the third phase, team proceeded with complicated platform launch and promotion, including a detailed launch plan developed with Swin HCM - Vietnam, potentially featuring a 'Beauty of Artisans' day. Specifically, team implemented culturally sensitive digital marketing campaigns across social media (Facebook, Zalo, Instagram), local online communities, and through collaborations with relevant influencers and tourism platforms, emphasizing the unique stories and fair trade principles behind each product. Furthermore, this suitable approach ensured a seamless experience and established clear customer support mechanisms via FAQs, contact forms, and potential live chat/phone support primarily in Vietnamese. As a result, the commitment extended to ongoing platform monitoring and maintenance, including regular security audits, bug fixes, updates, and performance tracking, with reports provided to Swin HCM - Vietnam to ensure the platform's long-term viability.

2.3.5 Fifth Phase

On the last phase, team determined that primary object is to facilitate the long-term self-management and sustainability of the e-commerce platforms by artisan communities. Consequently, this approach involved developing clear Community Management Guidelines in close collaboration with the artisans and their representative organizations, covering platform ownership, content

moderation, and dispute resolution to align with cultural protocols. Consequently, implementing a robust Knowledge Transfer program provided specialized training to designated community members on platform administration, advanced content management, and basic technical support. Finally, in partnership with local leadership, we will develop a comprehensive Sustainability Plan, outlining financial models like fair commission structures, ongoing maintenance strategies, and identifying external funding opportunities to ensure the project's lasting impact and foster a self-sufficient ecosystem for cultural heritage preservation and artisan livelihoods.

2.4 Conclusions and Recommendations

In conclusion, the team's efforts in general and my individual contributions in particular were pivotal in developing a culturally suitable e-commerce solution for traditional handicrafts from Southern Vietnam. Thus, my leadership ensured a focus on key design principles, notably the creation of user-friendly interfaces tailored for varying digital literacy levels and the integration of robust fair trade mechanisms. Additionally, these elements were central to our platform's design, ensuring it not only expanded market access but also respected and preserved the cultural heritage of the artisan communities. Therefore, I would advocate for more iterative and direct engagement with a broader spectrum of artisan representatives during the initial design phases by enhancing outstanding design ideas within a team environment. Consequently, this approach allows team members to refine features, ensure the platform remains maximally responsive to the evolving needs and cultural nuances of its users, thereby strengthening its overall impact and practical utility.

Part C

This section indicates my achievement of Unit Learning Outcomes (ULOs) principles, demonstrating how the project experience has enhanced my understanding and application of relevant skills and knowledge. Therefore, each ULO is addressed using the STAR (Situation-Task-Action-Response) framework, supported by specific examples from our team's activities and my individual contributions.

3.1 Locate Indigenous knowledge systems and consider how they story the long history of technology, science, and engineering.

Situation: The project's aims to initialize and develop e-commerce platforms for traditional handicrafts from Southern Vietnam which inherently involved engaging with communities that possess rich Indigenous knowledge systems regarding their crafts and the historical technologies used in their creation. Notably, main task of team was to understand these systems and ensure the developed solution respected and highlighted them.

Task: As the team leader, I launched a detailed plan where team members involved understanding and exploring Indigenous knowledge systems to ensure the project was undertaken in accordance with them during the project. Subsequently, this approach allows team members not only to go beyond simply digitizing products dive into the stories and traditional techniques behind each handicraft..

Action: Team GAMBLE launched initial research into the cultural significance and traditional production methods of various handicrafts from Southern Vietnam, recognizing that these crafts are embedded within the long history of technology and cultural expression of these communities and emphasizing ethical representation of artisans and their crafts in our project vision.

Result: The above approach offered team members to design a platform that not only facilitated economic empowerment but also served as a respectful repository and showcase for the unique cultural heritage and traditional technological practices embodied in the handicrafts, ensuring the project's alignment with Indigenous contexts.

3.2 Apply relevant knowledge of emerging technologies to a project within an Indigenous context taking into consideration and acknowledging Indigenous histories, worldviews, standpoints, and cultures.

Situation: The project outlined the bridge gap between traditional handicraft communities facing limited digital literacy and market reach, and the opportunities presented by modern e-commerce technologies. The challenge was to apply emerging e-commerce solutions in a way that was culturally appropriate and accessible.

Task: Beyond to the enthusiastic leadership of the team leader, team was able to apply knowledge of emerging technologies to create a user-friendly and culturally appropriate e-commerce platform that acknowledged the unique histories, worldviews, and cultures of the artisan communities which involved analyzing their challenges and needs

Action: Team GAMBLE used modern web development applications, tools, and techniques to build a robust and secure e-commerce platform. Notably, team members prioritized user-centered design, ensuring the platform's interface was intuitive for artisans with varying levels of digital literacy, a key consideration within this Indigenous context. Moreover, members within the team also figured out various digital marketing strategies including Facebook, Zalo, Instagram, and local online channels that could effectively reach target audiences while maintaining cultural sensitivity.

Result: Consequently, by carefully implementing, updating, and integrating user-friendly design and appropriate digital marketing strategies within a culturally sensitive framework, the project successfully applied emerging technologies to create a viable online marketplace that respects and benefits the artisan communities, ensuring accessibility and relevance for their unique context.

3.3 Function as an effective team member using project management tools and demonstrating professionalism and ethical behaviour.

Situation: The project criteria required a collaborative effort from a diverse team to address the multifaceted challenges of developing e-commerce platforms for artisan communities that effective teamwork, project management, and ethical conduct were paramount for success.

Task: As a team leader, my task was to contribute effectively, utilize project management tools, and maintain professionalism and ethical behavior throughout the project lifecycle. Specifically, Hieu was responsible for

organizing crucial meetings and tracking deadlines, ensuring team members keep up with workflow and maintain momentum.

Action: Team leader actively organized and participated in team meetings and workshops, contributing to the planning and execution of each phase. Therefore, all team members frequently utilized powerful tools like Google Docs for collaborative idea generation, note-taking, and real-time feedback, streamlining our workflow. Interestingly, I ensured my deliverables were submitted on time, supporting the team's overall progress. As a result, team collectively adhered to fair trade principles and ethical representation, ensuring the project's integrity.

Result: By using project management tools and consistent professional and ethical conduct, team GAMBLE successfully progressed through each phase, from initial research and design to the detailed planning of training and launch, demonstrating strong collaborative efficiency and a commitment to the project's values.

3.4 Communicate within teams, stakeholders using appropriate verbal, written, and technological approaches.

Situation: In today's world, effective collaboration was critical given the multi-faceted nature of the project, involving internal team members, the issuing organization (Swin HCM - Vietnam) , and ultimately the artisan communities themselves.

Task: As a position of team leader, my task was to communicate effectively within the team and with stakeholders using appropriate verbal, written, and technological approaches during the project. In addition, this approach included contributing to meetings, engaging with facilitators, and proficiently using various communication methods

Action: Based on requirements in each phase of the project, I actively contributed to team meetings, providing updates and engaging in discussions for various phases of the project. In written communication phase, I decided to write reports and take notes for online interactions, ensuring clarity and conciseness , as evidenced by the project documentation. In technologically phase, team leveraged collaborative platforms like Google Docs for real-time sharing and feedback , which significantly enhanced our ability to refine concepts and track progress. Finally, all team members also planned for communication with artisans to be primarily in Vietnamese, demonstrating a culturally sensitive approach.

Result: Surprisingly, the effective of using diverse communication channels and approaches facilitated seamless collaboration within the team and with Swin HCM - Vietnam, leading to a cohesive project development process and ensuring that crucial information was shared efficiently, which is vital for a project with such significant community engagement

